

Harmony — Personal Journal Auditing Web App

Google Cloud Gen AI Academy APAC 2026 Ideathon Challenge | Architecture & Security Deck

Live Prototype: <https://personal-gemini-journal-344135619629.asia-south1.run.app>

■ Codelab Compliance Checklist

Codelab Requirement	Implementation Detail	Status
1. AI Studio Custom Instructions	Enforced security system prompt covering threat modeling, UID isolation & sanitization.	■ Completed
2. Firebase Authentication	Federated Google Sign-In & Email/Password auth integrated via Firebase Client SDK.	■ Completed
3. Multi-turn AI Interaction	Conversational reflection engine using Gemini API with conversation history context.	■ Completed
4. User-Isolated Firestore	All records stored under /users/{userId}/* enforced by strict request.auth.uid == userId rules.	■ Completed
5. Secret Manager Integration	GEMINI_API_KEY fetched dynamically at runtime from GCP Secret Manager (zero hardcoded keys).	■ Completed
6. Cloud Run Hosting	Deployed with dedicated service account personal-gemini-journal-sa@ai-barista-track-1...	■ Completed
7. Mandatory Verification Label	Deployed with label --update-labels dev-tutorial=cloud-run-ai-challenge.	■ Completed
8. Original Feature Enhancements	Harmony Journal Insights analytics dashboard & AI-extracted Action Items checklist.	■ Completed

■ Security Verification Matrix

Test Scenario	Description	Expected Result	Status
Unauthenticated Route Block	Access /journal or /insights without Firebase Sign-In	Redirected to /login	■ Passed
Cross-User Firestore Read	User A queries /users/UserB_UID/interactions	Permission Denied by Security Rules	■ Passed
Cross-User Firestore Write	User A writes to /users/UserB_UID/action_items	Permission Denied by Security Rules	■ Passed
Secret Key Protection	Inspect browser network tabs and client JS bundles	GEMINI_API_KEY is 100% absent	■ Passed
Challenge Verification Tag	Inspect Cloud Run service metadata labels	dev-tutorial=cloud-run-ai-challenge present	■ Passed

■■ Architecture Overview

Cloud Run Hosting: Deployed as containerized Next.js 16 service with dedicated IAM service account (personal-gemini-journal-sa).

Authentication: Firebase Auth with Google Sign-In & Email/Password.

Database Isolation: Cloud Firestore document storage under /users/{userId}/* with strict security rules.

Secret Manager Integration: GEMINI_API_KEY fetched at runtime via Secret Manager API (zero hardcoded secrets).

AI Engine: Google Gemini API powering multi-turn reflection, sentiment trend scoring, task extraction & chart audits.